Scoring, substitution matrices and score aggregation for TCR antigen-specificity search

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vdjmatch technical appendix — draft, regenerated from `bench/gen_vdjam.py`, `bench/loo_vdjam.py`, `bench/gen_appendix_assets.py`

Abstract

`vdjmatch` annotates T-cell receptor (TCR) clonotypes by antigen specificity through fuzzy CDR3 search against a labelled database (VDJdb), built on the `seqtree` search core. This note documents the *scoring* layer: how CDR3 similarity is scored, the data-driven TCR substitution matrix (VDJAM) and why it is region-structured, and how per-hit scores are aggregated into a control-calibrated significance. CDR3 loops resist the assumptions behind BLOSUM/PAM — they cannot be multiply aligned across clonotypes of different length, and germline-encoded flanks (`CASS`...) are so over-conserved that a naive matrix conflates germline bias with selection. We give a position-debiased log-odds estimator that isolates the substitution signal, show empirically that this signal lives in the non-template (NDN) core while the V/J flanks are germline-determined, and demonstrate the consequences on worked examples (GIL, NLV) using the alignments and CIGAR strings produced by `seqtree`. The E-value theory used for significance is derived in the companion `seqtree` appendix; here we summarize it and give the paired-chain extension.

1. The search-and-score framework

Given a query CDR3 q and a labelled reference set D (VDJdb), `vdjmatch` retrieves the neighbours of q within a fixed scope $\theta = (s, i, d, t)$ (max substitutions, insertions, deletions, total edits) using the `seqtree` engine, which defines a ball $B_\theta(q) = \{x : \text{edit}(q, x) \leq \theta\}$ and returns, per hit, the edit decomposition and a matrix-weighted alignment. The matrix enters as a *ranking* score inside the ball; the ball itself (an edit budget) bounds the candidate set. Each hit carries its antigen labels, an extended CIGAR (=/`X`/`I`/`D`), and an alignment score. Significance comes from comparing the neighbour count to a matched background control (§ 3).

2. Substitution scoring and the VDJAM matrix

How BLOSUM is built, and why it does not transfer.. BLOSUM is a log-odds matrix estimated from the BLOCKS database of *ungapped multiple-alignment blocks* of conserved protein regions. Its recipe is: (i) take many trusted blocks — columns of homologous, gaplessly aligned

*Supplementary technical note for the `vdjmatch` software; intended as supplementary material to the main manuscript.

residues; (ii) cluster sequences within a block above a percent-identity threshold and down-weight within-cluster pairs so abundant near-duplicates do not dominate; (iii) count, over all blocks, how often residues a and b co-occur in the *same alignment column*, q_{ab}; (iv) score $s(a,b) = \frac{1}{\lambda}\log(q_{ab}/p_a p_b)$ against the *global* background frequencies p_a. The signal is "which residues substitute for one another within conserved, column-homologous positions," and the background is position-independent.

CDR3 loops break every step. (i) There is no trusted multiple alignment: same-specificity CDR3s differ in length and have no column homology, so "alignment columns" do not exist. (ii) The germline-encoded ends are over-conserved — essentially every human TRB CDR3 begins `CASS` and ends in a short J motif — so a *global* background p_a is wrong: an alanine in the N-terminal `CASS` is germline-fixed and carries no specificity signal, while an alanine in the centre does, yet BLOSUM would score them identically. A single global matrix estimated over all positions thus conflates germline conservation with antigen-driven selection.

What VDJAM changes (point by point).. VDJAM keeps the log-odds idea but replaces the two broken ingredients. (a) *Observation unit*: instead of columns of a multiple alignment, the unit is a *same-antigen single-substitution pair* — two CDR3s annotated to the same epitope, equal length, differing at exactly one position; this is the only "aligned residue pair" obtainable without a multiple alignment, and (like a BLOCKS column pair) it records two residues that are interchangeable without changing function (here, specificity). (b) *Background*: instead of a global p_a, a *position-specific* background $\mathrm{bg}(\cdot \mid L, p)$ (the residue frequency at that length and position) is used, so over-conserved columns are normalized away analytically rather than masked by hand. (c) *Redundancy*: BLOSUM's percent-identity clustering is replaced — partly by the position-specific background (germline-shared residues are "expected" and contribute little), and prospectively by the control-calibrated significance layer (§3) that already discounts convergent public clones; per-clonotype clustering can be added exactly as in BLOSUM if needed. (d) *Scope*: BLOSUM is one global matrix; VDJAM is estimated per chain and per region, because (next paragraph) only the NDN core carries a transferable substitution signal.

Observation unit and estimator.. We replace "alignment columns" with *same-antigen single-substitution pairs*: ordered pairs of CDR3s annotated to the same epitope, of equal length, differing at exactly one position — the only clean substitution signal extractable without a multiple alignment. For a substitution $a \leftrightarrow b$ observed at position p in a CDR3 of length L on chain c, let $\mathrm{bg}(\cdot \mid L, p)$ be the residue frequency at position p among all length-L chain-c CDR3s. With $f(a,b)$ the observed count of $\{a,b\}$ events and

$$e(a,b) = \sum_{\text{event slots } (L,p)} n_{L,p}\left[\mathrm{bg}(a \mid L,p)\,\mathrm{bg}(b \mid L,p) + \mathrm{bg}(b \mid L,p)\,\mathrm{bg}(a \mid L,p)\right], \qquad \mathrm{VDJAM}(a,b) = \log_2 \frac{f(a,b)+\alpha}{e(a,b)+\alpha}, \tag{1}$$

where $n_{L,p}$ is the number of observed events at (L,p) and α a Laplace pseudocount. The key point is that the background is *position-specific*: at an over-conserved column the residue frequency is peaked, so substitutions there are both rare (small f) and "expected" (matched e), and contribute ≈ 0 to the log-odds — the germline bias is removed analytically rather than by hand-masking. The matrix is consumed by `seqtree` via the squared-distance penalty $\mathrm{pen}(a,b) = s_{aa} + s_{bb} - 2s_{ab}$ (`SubstitutionMatrix.from_similarity`); a dominant uniform diagonal makes higher learned similarity rank a substitution better.

Region structure: NDN is learned, V/J are germline-determined.. A CDR3 is V-germline prefix · NDN insert · J-germline suffix. Estimating (1) separately by region shows (Fig. 1) that the substitution signal — the agreement with general amino-acid chemistry, measured as correlation with BLOSUM62 — is concentrated in the NDN core (TRB $r = 0.42$, TRA 0.35), whereas the V flank carries almost none (TRB 0.08, TRA 0.07): V-flank "substitutions" are dominated by germline allele differences, not free chemistry. This motivates a two-part model: **VDJAM is learned for the NDN core**, while the **V and J flanks are auto-computed** from the germline V/J sequences and their trimming. Concretely, a CDR3 position derived from V- (or J-) germline is near-invariant when it survives trimming and reverts to the random-insert background when trimmed; so the V/J contribution is a per-position conservation profile $w(p) = \Pr[\text{position } p \text{ is germline-retained}]$, determined by the gene's trimming-length distribution, rather than a substitution matrix learned from data. Combined, the NDN matrix scores the core and $w(p)$ down-weights substitutions at germline-retained flanks (realized through a per-position weighting of the base matrix). The germline-retention profile $w(p)$ is computed from the OLGA recombination model (per-gene V/J deletion distributions and germline CDR3-portion segment lengths, via `mirpy`'s `mir.basic.trimming`) and bundled with `vdjmatch`; Fig. 2 shows it drops monotonically from each anchor (the `CASS` V-stem and the J motif are germline; the core is insert). Region-aware scoring weights each CDR3 column by $1 - \Pr[\text{germline}]$ and applies the NDN matrix — implemented in `vdjmatch.match.regions` and evaluated below.

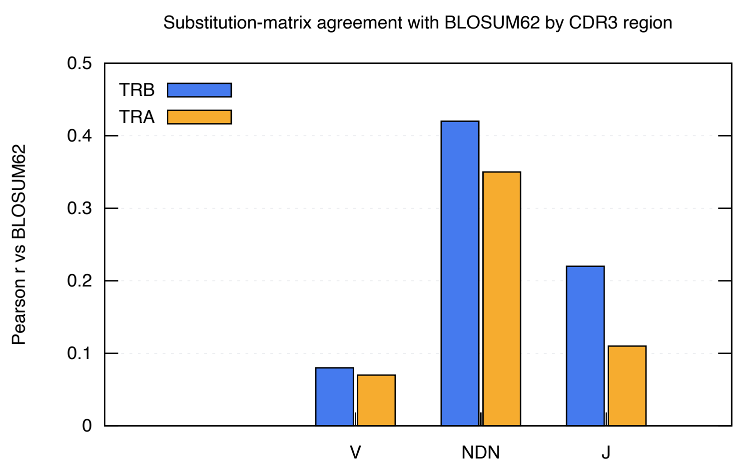

Figure 1: Per-region agreement of the learned CDR3 substitution matrix with BLOSUM62 (Pearson r of off-diagonal scores), human VDJdb. The free-substitution signal lives in the NDN core; the V flank is germline-determined.

3. Score aggregation and significance

Per-hit to per-query.. A region-aware similarity score sums the NDN matrix score over the core and the (down-weighted) flank contributions; this is available as an optional ranking score and

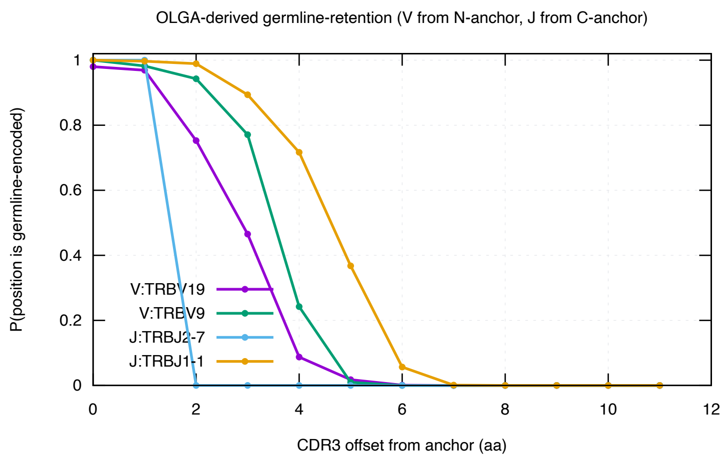

Figure 2: OLGA-derived germline-retention $w(p)$ (via `mirpy`): probability that each CDR3 offset from the V-Cys (or J) anchor is germline-encoded. It falls monotonically into the insert core, defining the region weight $1 - w(p)$ used in scoring.

generalizes the legacy cloglog generalized-linear aggregation over V/CDR1/CDR2/J/CDR3.

Control-calibrated E-value (primary).. Immune repertoires are biologically redundant, so an i.i.d. null over-calls. `vdjmatch` instead counts a query's VDJdb neighbours within the ball and compares to the count expected from a matched background control: with $N = |D|$, M the control size and n_C the control neighbour count, $\mathbb{E} = (N/M)\, n_C$ is the expected number of VDJdb neighbours by chance and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D)$ is the enrichment significance — significant only when a clonotype has *more* VDJdb neighbours than the generative background predicts. The Poisson/Chen–Stein theory, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix.

Paired $\alpha\beta$ E-value.. The null factorization is not an approximation of convenience: α and β chains pair *almost without constraint* in the repertoire at large, as shown by integrating structural and paired-sequencing data [1], so under the background (generative) law the joint ball mass genuinely factorizes, $\pi_0^{\alpha\beta} \approx \pi_0^{\alpha}\pi_0^{\beta}$. Crucially, the *same* analysis finds that *antigen-driven selection distorts* this uniform pairing landscape — i.e. conditioned on an epitope the two chains are *not* independent. This is exactly the structure the joint E-value exploits: the null factorizes (background pairing is unconstrained), so among the N paired VDJdb complexes the expected number of chance joint matches is $\mathbb{E}_{\alpha\beta} = N\, \pi_0^{\alpha}\pi_0^{\beta}$ with p the Poisson tail on the count of complexes matched in *both* chains; an excess over this factorized null is precisely the epitope-conditional dependence (co-selection of α and β) that signals a true specificity. Because the joint null is far smaller than either chain's, true pairs are dramatically more significant: on the paired TCRvdb benchmark, 296/300 pairs match a complex and 295/296 (99.7%) recover the correct epitope, with joint p down to 3.6×10^{-9}. (When the two chains' hits are treated as independent tests, the equivalent statistic is Fisher's combination of the per-chain enrichments; the appendix `seqtree` co-occupancy term quantifies the residual dependence.)

4. Results: does VDJAM (and region weighting) help?

A matrix derived from same-antigen pairs must be tested on epitopes it never saw, or the evaluation is circular. For each held-out epitope E^* we re-derive the NDN matrix from *all other* epitopes, fix the candidate set with a single unit-cost search on E^* (exact self excluded), and re-score each (query, hit) pair four ways — unit, BLOSUM62, NDN-VDJAM, and region-weighted NDN-VDJAM — ranking by score and reporting micro PR-AUC for "hit shares E^*" (Fig. 3, eight largest TRB epitopes). Two honest findings. (i) *Region weighting is a real, consistent, but small improvement over the flat matrix* (+0.004 at scope 2, +0.007 at scope 3; never worse on any epitope, and the gain grows with scope) — modest because CDR3 variation is *already* NDN-concentrated, so down-weighting the rarely-varying germline flanks moves few hits; it should matter more for cross-V-gene comparisons, where flank (germline-allele) differences are common. (ii) *BLOSUM62 remains the strongest substitution matrix at this narrow-scope retrieval task* (mean PR-AUC 0.32 vs 0.28 region-VDJAM vs 0.29 unit): the data-derived NDN matrix captures general substitution tolerance that BLOSUM already encodes well, and with limited same-antigen pairs it is noisier. At scope 2 the search ball supplies most of the specific-vs-nonspecific discrimination and any matrix only re-ranks within it; the substitution model is therefore a second-order lever here, and the first-order signal is the control-calibrated neighbour count (§ 3).

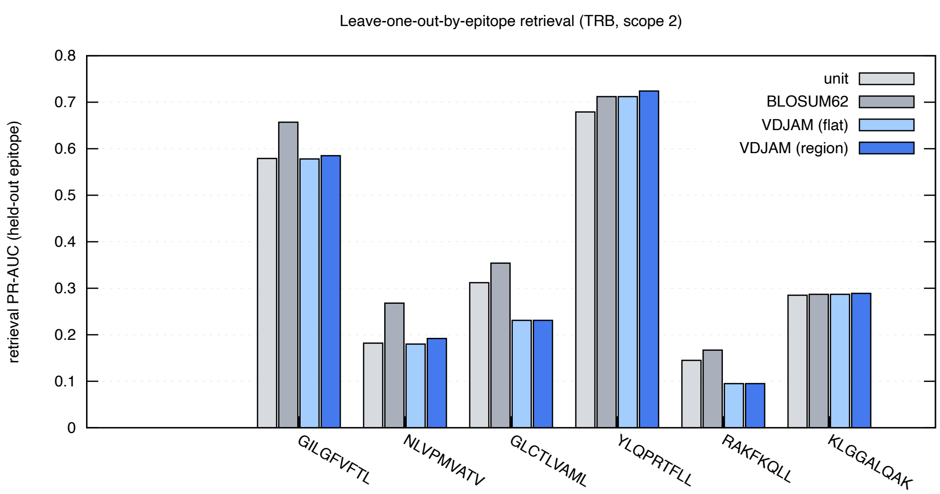

Figure 3: Leave-one-out-by-epitope retrieval PR-AUC (TRB, scope 2): the NDN VDJAM matrix versus BLOSUM62 and unit cost on epitopes excluded from matrix estimation.

5. Edit distance, position, and the matrix comparison

Four analyses (`bench/scoring_analysis.py`, `bench/loo_vdjam.py`) settle what scoring actually buys on VDJdb (human TRB).

Hamming distance 1 is the signal:noise optimum.. The fraction of CDR3 neighbours that share an epitope — the precision of a similarity call — falls steeply with edit distance: 0.53, 0.27, 0.19, 0.15, 0.13

at distance $1 \ldots 5$ (Fig. 5). A distance-1 neighbour is twice as likely to be a true co-specific as a distance-2 one and four times a distance-5 one. This reproduces the original VDJdb observation [3] that single-mismatch neighbours are the sweet spot, and it frames everything below: the search ball, not the substitution matrix, is the dominant lever; the matrix only re-ranks within a ball whose purity is already set by its radius.

Central substitutions carry the specificity signal.. Conditioning on *where* a single substitution falls (Fig. 6), the probability that the two CDR3s still share an epitope dips to ≈ 0.41 in the NDN core but rises to ≈ 0.70 toward the J border: a central mismatch most often *changes* specificity (it is contact-proximal and informative), while a border mismatch is usually germline noise (the conserved V/J stems). This is the empirical basis for weighting the centre more — the same direction as the germline-retention weight $1 - w(p)$ (§ 2) — and, encoded directly as a positional weight on BLOSUM62 (the PSSM of § 5), it is the one change that consistently improves retrieval over flat BLOSUM62 (Table 1). It also matches the contacting-region picture of Shcherbinin et al. [2].

The NDN core is glycine-rich.. Central CDR3 positions reach $\sim$ 30–36% glycine in both chains (Fig. 7) — the N-region/D-gene insertion signature — a strongly non-BLOSUM composition. Notably TRA is comparably central-G-rich to TRB, so this is an insertion-diversity feature, not a TRB-D-gene-exclusive one; either way the NDN alphabet is distinctive enough that a general matrix is a poor fit there in principle.

No *standard* matrix beats BLOSUM62.. Re-scoring the held-out-epitope candidates (leave-one-out, § above) with each substitution matrix gives the same verdict at every scope (Table 1, Fig. 4): BLOSUM62 and PAM250 are the best and essentially tied, the structural matrix sits between them and plain Hamming, Hamming (unit) is a strong baseline, and the data-derived NDN-VDJAM — flat or region-weighted — trails them all (region beats flat by a small, consistent, scope-growing margin). So among off-the-shelf and data-derived *amino-acid* matrices, the honest answer to "can we beat BLOSUM62" is no: the substitution matrix is a second-order lever once the ball radius is fixed, and the first-order statistic remains the control-calibrated neighbour count at distance 1 (§ 3).

But position-weighting BLOSUM62 does (the significance PSSM).. Experiment 2 says *where* a substitution falls matters more than *which* substitution it is. We encode that directly as a positional factor on top of BLOSUM62 rather than as a new alphabet matrix. From the position-significance curve (Fig. 6) we take a per-relative-position weight $\omega(p) = (1 - \Pr(\text{same} \mid \text{sub at } p))$, normalised to mean 1 (centre ≈ 1.4, J border ≈ 0.7), and interpolate it to any CDR3 length. `seqtree` consumes this natively: `regions.significance_pssm()` builds a fixed-width `PositionalMatrix.from_weights(BLOSUM62,` $\lfloor 100\,\omega(p) \rceil$`)`, so the engine scores $\text{pen}(p, a, b) = \omega(p) \cdot \text{BLOSUM62}(a, b)$ — a central mismatch costs $\sim 1.4\times$ its border-position cost. This is the only scoring change that beats flat BLOSUM62, and it does so *at every held-out epitope*: $0.333 \to 0.356$ at distance ≤ 2 and $0.218 \to 0.236$ at distance ≤ 4, winning 8/8 epitopes at both radii (Table 1). The gain is modest in absolute PR-AUC but uniformly signed, exactly because the matrix re-ranks within a ball whose purity is already set by the radius. The remaining first-order lever is the *V-gene* dimension (cross-V comparisons and V-similarity grouping), where germline-flank differences are common and this central weighting should matter most.

mean retrieval PR-AUC	unit	BLOSUM62	PAM250	structural	VDJAM	VDJAM-region	BLOSUM+possig
edit distance ≤ 2	0.306	0.333	0.329	0.314	0.284	0.287	**0.356**
edit distance ≤ 4	0.203	0.218	0.218	0.192	0.177	0.181	**0.236**

Table 1: Substitution matrices compared by leave-one-out-by-epitope VDJdb-vs-VDJdb retrieval (TRB, 8 largest epitopes, micro PR-AUC). Among standard/data-derived amino-acid matrices BLOSUM62 $\approx$ PAM250 lead and VDJAM trails; the only scoring that beats BLOSUM62 is BLOSUM62 reweighted by the positional-significance PSSM (centre $>$ V/J borders), which wins all 8/8 held-out epitopes at both radii.

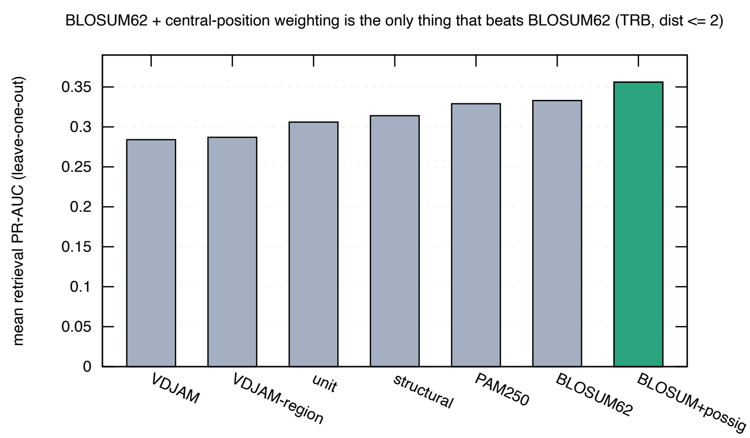

Figure 4: Mean leave-one-out retrieval PR-AUC by scoring (TRB, distance $\leq$ 2). No standard amino-acid matrix beats BLOSUM62; reweighting BLOSUM62 by the central-position significance factor (green) does.

6. Worked examples

Pairwise (different lengths).. `seqtree` aligns CDR3s of differing length, placing indels where the loop genuinely lengthens. The edit-distance alignments below (influenza GIL-specific TRB CDR3s) show the conserved `CASS` V-anchor and `DTQYF` J-motif with substitutions and indels confined to the central NDN core:

```
query (ref): CASSPRSGETQYF   epitope GILGFVFTL   (unit / edit-distance alignment)

vs CASSNRSGETQYF  (len 13; subs=1 ins=0 dels=0; CIGAR 4=1X8=)
  CASSPRSGETQYF
  |||| ||||||||
  CASSNRSGETQYF

vs CASSIRSGESTQYF  (len 14; subs=1 ins=1 dels=0; CIGAR 4=1X4=1I4=)
  CASSPRSGE-TQYF
  |||| |||| ||||
  CASSIRSGESTQYF
```

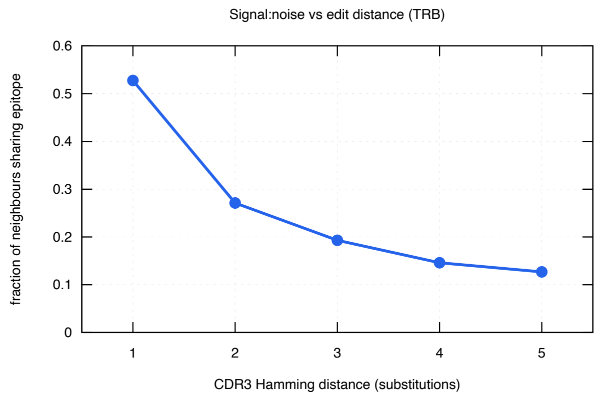

Figure 5: Fraction of CDR3 neighbours sharing an epitope vs Hamming distance (TRB): distance 1 is the signal:noise optimum [3].

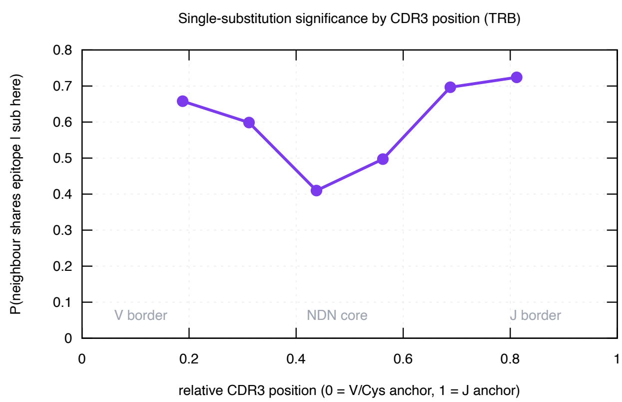

Figure 6: Pr(neighbour shares epitope | single substitution at relative position), TRB. The NDN core is where a mismatch most changes specificity; the V/J borders are germline noise.

```
vs CASSLFSETQYF  (len 12; subs=2 ins=0 dels=1; CIGAR 4=2X1=1D5=)
  CASSPRSGETQYF
  ||||  | |||||
  CASSLFS-ETQYF

vs CASSLGETQYF  (len 11; subs=1 ins=0 dels=2; CIGAR 4=2D1X6=)
  CASSPRSGETQYF
  ||||   ||||||
  CASS--LGETQYF

vs CASSPWSGSQETQYF  (len 15; subs=1 ins=2 dels=0; CIGAR 5=1X2=2I5=)
  CASSPRSG--ETQYF
  ||||| ||  |||||
  CASSPWSGSQETQYF
```

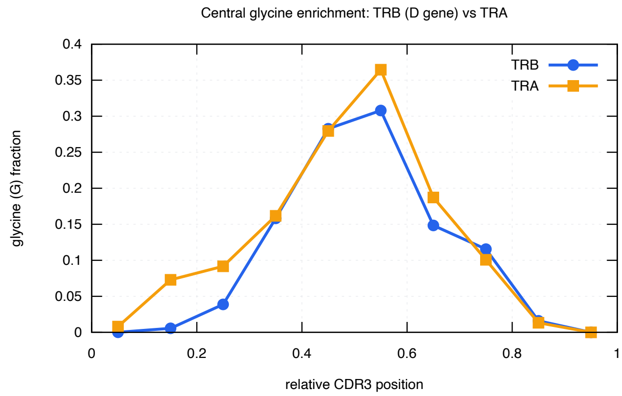

Figure 7: Glycine fraction by relative CDR3 position: the NDN core is $\sim$ 30–36% glycine in both TRA and TRB (insertion/D-gene signature).

Multiple (same length).. Aligning same-length same-epitope CDR3s (no indels needed) makes the conservation profile explicit (Figs. 8–9): columns are shaded by conservation, dark at the germline-determined `CASS` prefix and J motif, light through the variable NDN core. This is the structural counterpart of the finding of Shcherbinin et al. [2] that specificity-preserving CDR3 substitutions occur predominantly outside the peptide/MHC-contacting positions while preserving the loop backbone — i.e. the tolerated variation we score in the NDN core is exactly the variation that does not break recognition.

7. Limitations and roadmap

The data-derived NDN matrix is estimated from a limited set of same-antigen single-substitution pairs and does not beat BLOSUM62 at any scope; region-aware germline+trimming weighting gives a small, consistent gain over the flat matrix but not over BLOSUM62. The substitution *alphabet* is thus a second-order lever. What does move the needle is *position*: reweighting BLOSUM62 by the central-significance PSSM beats flat BLOSUM62 at every held-out epitope (§ 5), confirming that for CDR3 *where* a mismatch falls matters more than *which* residue it is. The clearest remaining lever is the *V-gene* dimension: a V-gene similarity term (grouping V genes by CDR1/CDR2/CDR2.5 distance) and cross-V comparisons, where germline-flank differences are common and this position weighting should matter most; a per-chain treatment (TRA vs TRB) is also pending. Separately, the control's V–J usage must match the query repertoire's or the null is mis-specified — whether to renormalize V–J usage (at the risk of erasing genuine V-gene bias) is an open question to settle on held-out epitopes. These extensions, and the full benchmark against existing tools, are ongoing work.

References

[1] Dmitrii S. Shcherbinin, Vlad A. Belousov, and Mikhail Shugay. Comprehensive analysis of structural and sequencing data reveals almost unconstrained chain pairing in TCR$\alpha\beta$ complex. *PLoS Computational Biology*, 16(3):e1007714, 2020.

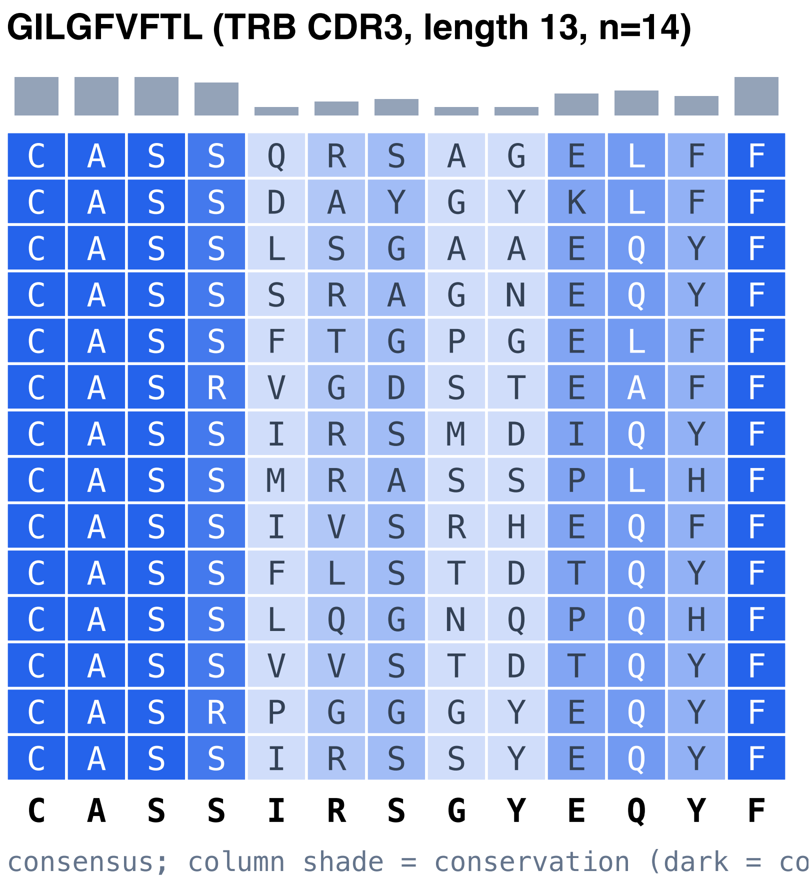

Figure 8: GIL (influenza M1, A*02) TRB CDR3 block; conserved V/J flanks, variable NDN core.

[2] Dmitrii S. Shcherbinin, Vadim K. Karnaukhov, Ivan V. Zvyagin, Dmitriy M. Chudakov, and Mikhail Shugay. Large-scale template-based structural modeling of T-cell receptors with known antigen specificity reveals complementarity features. *Frontiers in Immunology*, 14:1224969, 2023.

[3] Mikhail Shugay, Dmitriy V. Bagaev, Ivan V. Zvyagin, Renske M. Vroomans, Jeremy Chase Crawford, Garry Dolton, et al. VDJdb: a curated database of T-cell receptor sequences with known antigen specificity. *Nucleic Acids Research*, 46(D1):D419–D427, 2018.

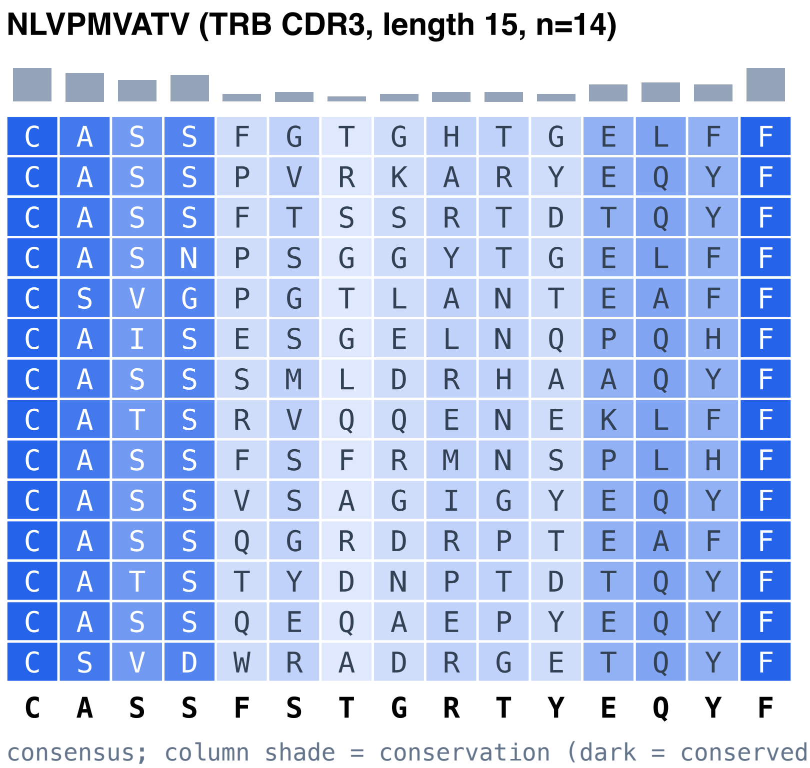

Figure 9: NLV (CMV pp65, A*02) TRB CDR3 block; same conservation pattern.